PRODUCT SAFETY DATASHEET

Lithium ion nickel manganese cobalt oxide battery

NTBA0V0108AA – N18650CNP

According Regulation (EC) No 1907/2006 (REACH) a safety data sheet must be provided for substances and preparations only. Batteries are not affected by the requirements of this Regulation.

1 Identification of the substance and of the manufacturer

Identification:	Product category	Lithium Ion battery
	Model name	NTBA0V0108AA
	Nominal voltage	345.6 V
	Nominal capacity	30 Ah
	Nominal energy	10 368 Wh
	Chemical system	Lithium NMC
	Rechargeable	Yes
Manufacturer:	Name and address	Novum Tech ZI Villard Bozon ZA du BACON 38570 Goncelin

2 Hazards identification

Hazards: Lithium-ion batteries may present a risk of fire or explosion or chemical burn when mistreated. Do not short circuit, puncture, incinerate, crush, immerse, force discharge or expose to temperatures above the declared operating temperature range of the product. Read instructions carefully.

Under normal conditions of use, the chemicals and metals are contained in a sealed can and are not exposed to the outside. Risk of exposure only in case of abuse (mechanical, thermal, electrical) which leads to the activation of safety valves and/or the rupture of the battery housing. Electrolyte leakage, electrode materials reaction with moisture/water or battery vent/fire/explosion may follow, depending on the circumstances.

Touching live parts may cause electrical shock which may result in thermal heating or muscle paralyzing effects. The latter may cause ventricular fibrillation, sudden cardiac arrest or respiratory paralysis with fatal ending.

Toxicity: If a battery burns, the vapors can irritate eyes, skin and the respiratory tract.

3 Reaction equation, composition and information on ingredients

Oxidation: $Li_xC_6 \rightarrow Li_{1-x}C_6 + x\ Li^+ + x\ e^-$

Reduction: $Li_{1-x}Ni_aMn_bCo_cO_2 + x\ Li^+ + x\ e^- \rightarrow LiNi_aMn_bCo_cO_2$

The following components are found inside the sealed Li-ion battery can.

Component	Chemical name	CAS number
Lithium transition metal oxide[1]	Lithium cobalt oxide	12190-79-3
Carbon（graphite）	C	7782-42-5
Electrolyte	C3H4O3 C4H8O3 C3H6O3 F6LiP	96-49-1 623-53-0 616-38-6 21324-40-3
Copper	Cu	7440-50-8

[1] M means a combination of Co, Ni or Mn. This component may consist of a mixture of compounds, each of which may contain these elements.

Aluminium	Al	7429-90-5
Others		Various

4 First aid measures

The chemicals are contained in sealed cans. Upon normal conditions of use, risk of exposure occurs only if the battery is mechanically, thermally or electrically abused.

If chemicals leak attends these advices:

Inhalation: Contents of an opened battery can cause respiratory irritation. Provide fresh air and call a doctor.

Skin contact: Contents of an opened battery can cause skin irritation. Wash skin with soap and water.

Eye contact: Contents of an opened battery can cause eye irritation.

Immediately flush eyes thoroughly with water for 15 minutes and seek medical attention.

If it should have come to an electric shock, your *acting should be based on the following:*

- *do not touch the injured person until you have ensured the absence of voltage;*
- *take away exposed live cables from the injured person by using non-conductive items;*
- *primary objective in the treatment of unconscious patients is the maintenance of their breathing and cardiovascular system. If necessary, you have to give cardiopulmonary resuscitation.*
- *cool burn injuries and cover them with an aseptic and non-fluffy wound dressing.*

5 Fire Fighting measures

In case of fire: Use dry chemical extinguishers.

Caution: Before starting to extinguish the fire, be sure, that you are at windward of fire. So you cannot inhale toxic vapors.

6 Accidental release measures

Wipe up leaked electrolyte fluid with an absorbent cloth.

If there is a lot of leaked electrolyte, you should wear:

- protective clothing
- gas mask for organic gases
- safety goggles
- gauntlets

7 Handling and storage

Handling: Do not open the battery. Do not crush, disassemble, drop or solder.

Charging: Charge within limits of 0 °C to +50 °C temperature
Charge only with specified charger designed for this battery.

Discharging: Discharge within limits of -20 °C to +75 °C temperature.

Caution: Wrong handling can cause fire or explosion.

Storage: Temperature: -20 °C to +45°C

Humidity range: 0 % to 60 %

Well ventilated area.

Short circuit can ransom burn.

Do not store with metal objects.

8 Exposure controls and personal protection

Advices under Point 7 must be observed. You have to check continuously that storage temperature is within the specified limits. Additionally, you have to ensure that there is a suitable ventilation in order to avoid water logging.

For normal use you don't need any protective equipment.

9 Physical and Chemical Properties

Form: Solid;

Color: Various;

Odor: Odorless;

pH: Not available;

Flash point: Not available;

Flammability: Not available;

Vapor pressure: Not available;

Solubility (water): Insoluble;

10 Stability and reactivity

Cells are stable under normal conditions. The following substance may appear in case of fire or leakage: organic carbonate, hydrogen fluoride, carbon monoxide, carbon dioxide, phosphorus fluoride.

11 Toxicological information

Cells are not hazardous when used properly. In case of fire or leakage combustion and decomposition products may cause irritation and toxicity to skin, eye and respiratory systems.

Toxicity data of some substance is listed:

Hydrogen fluoride:

Extremely toxic, May be fatal if inhaled or ingested. Readily absorbed through the skin contact may be fatal. Possible mutagen. LCLO: 50 ppm/30m (human beings), LC50： 1276 ppm/1h (rats).

Carbon and graphite:

Slightly hazards in case of skin contact (irritant), ingestion, inhalation, which will cause chronic damage to upper respiratory tract and cardiovascular system.

Copper:

Dust may cause respiratory irritation.

LD50: 3.5 mg kg-1(mouse).

12 Ecological information

Upon normal use there won't be any environmental pollution. If the battery is unusable, you must recycle it. See Point 13.

13 Disposal considerations

The battery is hazardous waste.

It is not allowed to dispose it with common waste.

If the battery is unusable, dispose it according to the applicable recycling regulations.

14 Transportation information

14.1 UN number

UN 3480

14.2 UN proper shipping name

UN 3480:

ADR:	LITHIUM ION BATTERIES
ICAO-TI/IATA-DGR:	Lithium ion batteries

14.3 Transport hazard class

9 (*Miscellaneous*)
Section : IB

14.4 Packing group

II (*Substances presenting medium danger*)

Packing Instruction : 965

Note:

Cells and batteries, cells and batteries contained in equipment, or cells and batteries packed with equipment, containing lithium in any form shall be assigned to UN numbers 3090, 3091, 3480 or 3481 as appropriate. They may be carried under these entries if they meet the following provisions:

a. each cell or battery is of the type proved to meet the requirements of each test of the Manual of Test and Criteria, Part III, sub-section 38.3;

b. each cell and battery incorporate a safety venting device or is designed to preclude a violent rupture under normal conditions of carriage;

c. each cell and battery is equipped with an effective means of preventing external short circuits;

d. each battery containing cells or series of cells connected in parallel is equipped with effective means as necessary to prevent dangerous reverse current flow (e.g. diodes, fuses, etc.)

15 Regulatory Information

For shipping regulations see section 14.

16 Other information

This information has been compiled accurately to the best of our knowledge and belief. However, Novum Tech excludes any warranty for the accuracy, reliability or completeness of the information contained herein.

It is the user´s responsibility to satisfy himself as to the suitability and completeness of this information for his particular use.